Forecasting Omaha Temperatures Using Recurrent Neural Networks

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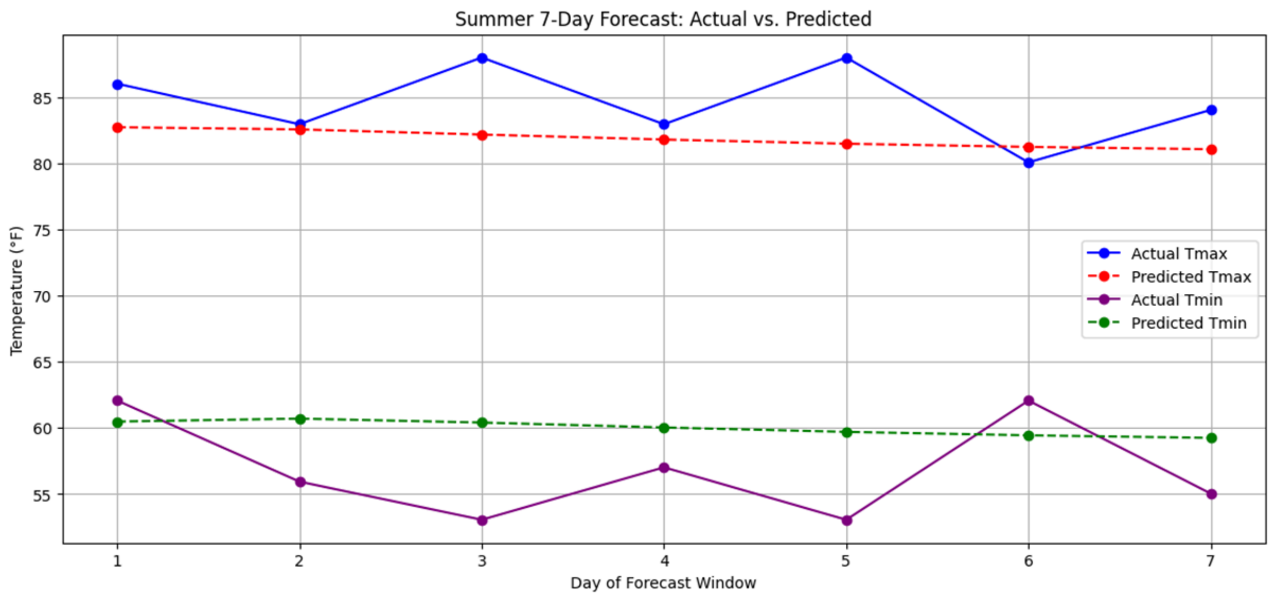
Accurate short-term temperature prediction is critical for applications related to agriculture, energy planning, transportation, and general environmental decision-making. While numerical weather prediction remains the foundation of operational forecasting, machine learning provides an appealing complementary approach in cases where computational cost is a concern or when local-scale features can be learned directly from historical data. The purpose of this project is to develop a machine-learning model capable of forecasting seven consecutive days of maximum and minimum temperatures with at least 70% accuracy during all four meteorological seasons. The overall goal is to evaluate whether a lightweight LSTM architecture, applied with recursive forecasting, can produce skillful predictions across a full seasonal cycle.

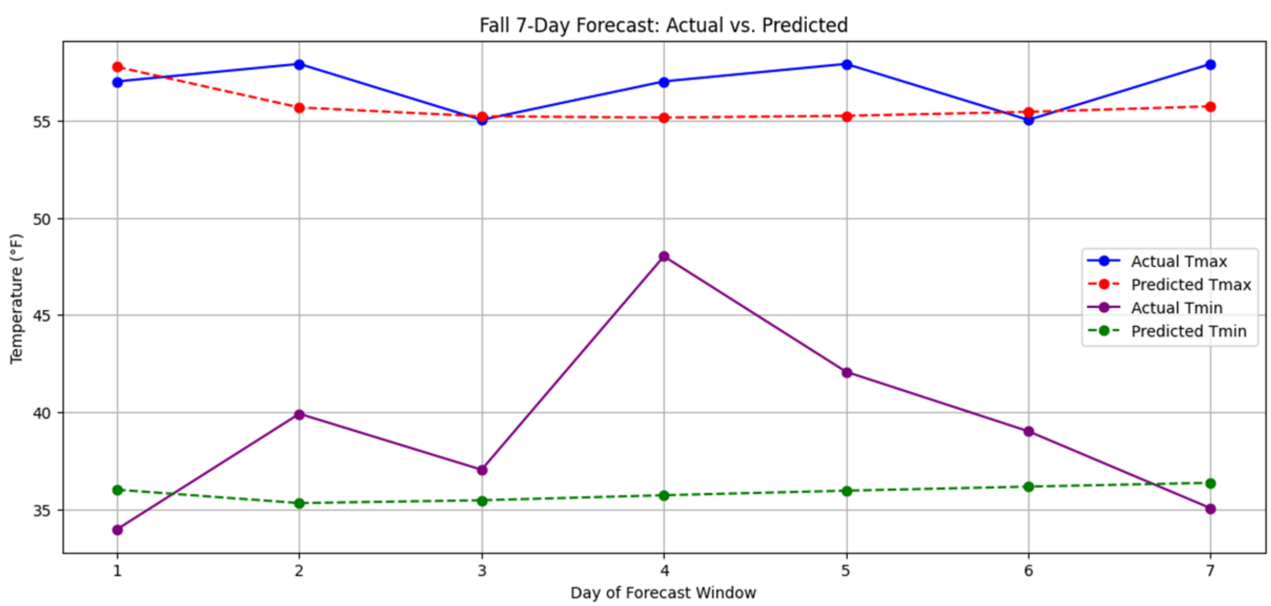
Daily weather observations from Omaha Eppley Airfield between January 1, 2016 and January 1, 2025 provide the basis for this study. The dataset includes maximum and minimum temperatures, average temperature, precipitation, snowfall, wind speed, and pressure. After removing variables with extensive missing values, temperatures are converted from Celsius to Fahrenheit, and additional predictors are engineered to capture relevant physical behavior. These include the diurnal temperature range and lagged values of the maximum and minimum temperatures from one and two days prior. These lag features help the model understand persistence, which is one of the strongest predictors in short-term temperature forecasting. The dataset is normalized using separate MinMax scalers: one for all predictor features and another for the temperature targets, which avoids any risk of scaling leakage. To reflect the strongly seasonal nature of temperature patterns, the dataset is partitioned by meteorological season. The model trained for winter only uses data from December through February; similarly, the spring, summer, and fall models use only their respective seasonal subsets.

To construct inputs for the LSTM, the model uses a sliding window of recent days to predict the next day's temperatures. Depending on the season, training windows between 30 and 45 days are used; longer windows help the model capture slower-evolving temperature regimes such as persistent cold spells in winter or prolonged heat waves in summer. Training sequences include only days before the selected 2024 forecast window. For each season, a seven-day evaluation period is identified in 2024: April 1-7 for spring, June 3-9 for summer, November 6-12 for fall, and December 25-31 for winter. These periods were chosen to be representative and climatologically stable relative to their seasonal norms. None of these dates are included in the training data, ensuring that the evaluation is a true forecast rather than a test of memorization.

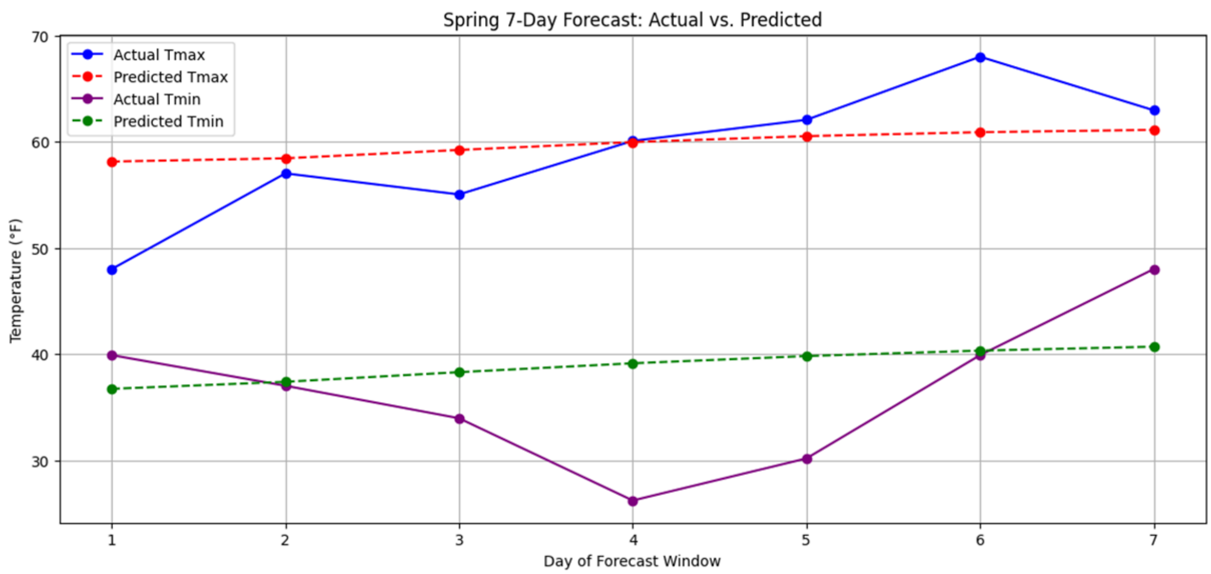
The forecasting model is a simple one-day LSTM. It takes the most recent sequence as input and predicts the next day's minimum and maximum temperatures. The architecture consists of a single LSTM layer with 64 units, a 32-unit dense layer with ReLU activation, and a final two-unit output layer corresponding to the predicted daily minimum and maximum. The model uses the Adam optimizer with mean absolute error as the loss function. Although this LSTM predicts only one day ahead, it can be extended to a seven-day forecast by recursively feeding each prediction back into the input window, updating the most recent timestep with the model output, and repeating the process seven times. This recursive strategy is lightweight and flexible, but also introduces error growth from day to day, addressing an obstacle in multi-step machine learning forecasting.

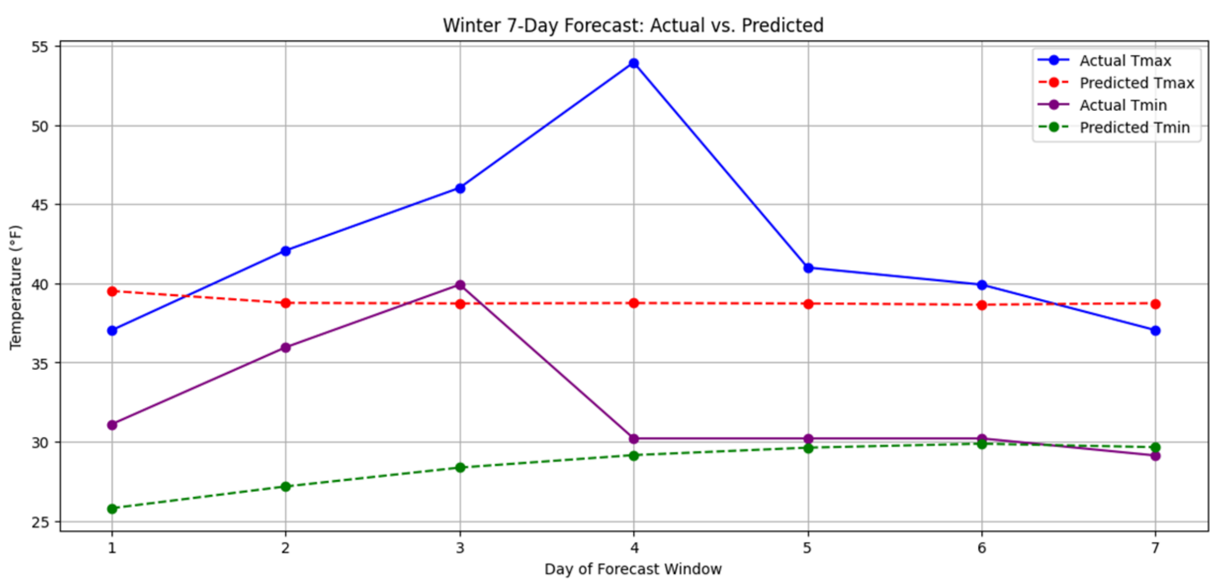
Model performance is evaluated using a relative accuracy metric derived from the mean absolute error. The scale is set at 20°F to represent typical daily temperature variability; values are restricted to be non-negative. This metric provides an intuitive sense of predictive skill: an accuracy near 1 corresponds to very small errors, while values near 0 indicate poor performance.





The model's performance varied across seasons, reflecting the different levels of temperature variability inherent to each forecast period. The LSTM-based forecasting system achieved its strongest results during the summer and fall intervals, where daily temperatures tend to evolve more smoothly and contain stronger autoregressive structure. For the summer forecast window (June 3-9), the model reached an overall accuracy of 84%, driven largely by an 88% accuracy in maximum temperature prediction and 80% accuracy for minimum temperatures. Similarly high performance was observed during the fall period (November 6–12), where the model achieved the highest overall seasonal score of 85%, including a peak 93% accuracy in maximum temperature forecasting and 77% accuracy for minimum temperatures. Loss curves for both seasons generally demonstrated good convergence within only a few epochs, reflecting stable gradients and well-behaved temporal patterns.





In contrast, the model exhibited moderate performance during the spring and winter forecast periods, which are climatologically more variable and contain sharper day-to-day swings. For the spring period (April 1-7), prediction accuracy reached 75% overall, with 79% accuracy for maximum temperatures and 71% for minimum temperatures. The winter forecast (December 25-31) was the most challenging, yielding 74% overall accuracy with nearly balanced performance between maximum (73%) and minimum (75%) temperature prediction. These weaker results correspond with the visual inspection of loss curves, which generally show slower convergence and higher validation error during the colder months. The winter plots in particular reveal that the model tends to underreact to abrupt warming or cooling events—consistent with greater atmospheric variability, frontal passages, and nonlinear transitions common in late December in the central United States.

There are also evident limitations. The dataset includes only surface variables from a single location, leaving out critical atmospheric predictors such as 850-mb temperature, geopotential height patterns, humidity profiles, or cloud cover. Incorporating reanalysis data would likely yield substantial improvements, particularly for minimum temperature. In addition, the training sets are relatively small after dividing by season. Future work could explore regional pooling, such as combining neighboring stations, to increase sample size without diluting climate signals. More advanced architectures, including sequence-to-sequence models or transformers, could directly predict all seven days at once, reducing error accumulation inherent in recursive approaches. Ensemble methods or probabilistic outputs could also provide uncertainty estimates valuable in operational contexts.

Despite these limitations, the project demonstrates that a simple one-day LSTM, when applied thoughtfully with season filtering, feature engineering, and recursive prediction, can achieve moderate-to-high skill for short-range temperature forecasting. Maximum temperatures in particular display strong predictability in summer and fall, while winter forecasts capture general variability patterns even if nighttime lows remain challenging. The findings show that machine learning can offer a computationally efficient supplementary tool for local-scale temperature prediction, and that seasonal partitioning meaningfully enhances model skill. With expanded datasets and improved architectures, the approach can be extended to more complex forecasting tasks and potentially integrated with traditional weather prediction systems.

References

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